



Dayananda Sagar Academy of Technology & Management

(Autonomous Institute under VTU)

Semester	:	3			
Course Title	:	ESSENTIALS OF DIGITAL CONNECTIVITY			
Course Code	:	23CSE35			
Course Type (Theory/ Practical/ Integrated)	:	Theory			
Category	:	PCC			
Stream	:	CSE	CIE	:	50 Marks
Teaching hours/ week (L:T:P:S)	:	3:0:0:0	SEE	:	50 Marks
Total Hours	:	40	SEE	:	3 Hours
Credits	:	3	Duration	:	

Course Learning Objectives: Students will be able to:

Sl. No	Course Objectives
1	To learn the basics of data communication networks and components.
2	To learn different layers of networks and protocols used in those layers.
3	To learn network requirements used for communication.
4	To learn to handle networking challenges by using different concepts
5	To learn the different approaches used for networking

Teaching-Learning Process

Pedagogical Initiatives:

Some sample strategies to accelerate the attainment of various course outcomes are listed below:

- Adopt different teaching methods to attain the course outcomes.
- Include videos to demonstrate various concepts in C.
- Encourage collaborative (Group) Learning to encourage team building.
- Ask at least three **HOTS (Higher-order Thinking Skills)** module-wise questions to promote critical thinking.
- Adopt **Problem-Based Learning (PBL)**, which fosters students' analytical skills, and develops thinking skills such as evaluating, generalizing, and analyzing information rather than simply recalling it.
- Show different ways to solve a problem & encourage the students to come up with creative & optimal-solutions
- Discuss various case studies to map with real-world scenarios and improve the understanding.
- Devise innovative pedagogy to improve **Teaching-Learning Process (TLP)**.



DSATM

Scheme of Teaching and Examinations for BE Programme -2024-25
Outcome Based Education and Choice Based Credit System (CBCS)
(Effective from the Academic Year 2024-25)

COURSE CURRICULUM

Module No.	Topics	Hours
1	Basics of Networking: Data Communication, Networks, Protocols, OSI Model, TCP/IP Protocol suite. Application Layer: Principles of Network Applications: Network Application Architectures, Processes Communicating, Transport Services Available to Applications, Transport Services Provided by the Internet, Application-Layer Protocols. The Web and HTTP: Overview of HTTP, Non-persistent and Persistent Connections, HTTP Message Format, User-Server Interaction: Cookies, Web Caching. T2: Ch 1: 1.1,1.2,1.4 Ch 2: 2.1-2.4, T1: Ch:2	8
Pedagogy	Demonstration, Think Pair and Share	
2	Application Layer: DNS; The Internet's Directory Service: Services Provided by DNS, Overview of How DNS Works, DNS Records and Messages, Socket Programming: creating Network Applications: Socket Programming with UDP, Socket Programming with TCP. Transport Layer 1: Introduction and Transport-Layer Services: Relationship Between Transport and Network Layers, Overview of the Transport Layer in the Internet, Multiplexing and Demultiplexing: Connectionless Transport: UDP, UDP Segment Structure, UDP Checksum, Principles of Reliable Data Transfer: Building a Reliable Data Transfer Protocol, Pipelined Reliable Data Transfer Protocols, Go-Back-N, Selective repeat. T1: Ch:2, T3: Ch 11, T1: Ch 3.	8
Pedagogy	Demonstration, Poster Presentation	
3	Transport Layer 2: Data Transfer, Flow Control, TCP Connection Management, Principles of Congestion Control: The Causes and the Costs of Congestion, Approaches to Congestion Control, TCP Congestion Control. The Network layer: What's Inside a Router?: Input Processing, Switching, Output Processing, The Internet Protocol (IP): IPv4, Addressing, IPv6, and More IPv4 Datagram Format IPv4 Datagram Fragmentation IPv4 Addressing Network Address Translation (NAT) ,IPv6. T1:Ch 3,Ch 4: 4.2,4.3	8
Pedagogy	Demonstration, Discussion	
4	Routing Algorithms: The Link-State (LS) Routing Algorithm, The Distance-Vector (DV) Routing Algorithm, Intra-AS Routing in the Internet: OSPF, Routing Among the ISPs: BGP, The Role of BGP ,Advertising BGP Route Information ,Determining the Best Routes, ICMP: The Internet Control Message Protocol. Network Management and SNMP: The Network Management Framework, The Simple Network Management Protocol (SNMP). T1:Ch 5: 5.2,5.4,5.6,5.7	8
Pedagogy	Problem Solving, Think Pair and Share	

5	Link Layer: Introduction to the Link Layer, The Services Provided by the Link Layer, Error-Detection and Correction Techniques, Parity Checks, Check summing Methods, Cyclic Redundancy Check (CRC), Switched Local Area Networks, Link-Layer Addressing and ARP, Ethernet. Physical Layer: Analog and Digital, Periodic Analog Signals, Digital Signals, Transmission Impairment, Data rate limits, Performance. T1:Ch 6: 6.1,6.2,6.4,T2: Ch 3	8
	Pedagogical Initiatives (Not limited to): • Think Pair and Share (Blended Learning): provides an opportunity for students to learn from one another. • Problem Solving: encourages cognitive thinking and enables creative problem solving. • Poster Presentation: allows students to represent the concepts visually to understand the topics easily. • Case studies: maps different domains in real time applications. • Demonstration: exhibits the implementation process.	

Textbooks	
Sl. No.	Title of the Book/Name of the author/Name of the publisher/Edition and Year
1	James F Kurose and Keith W Ross, Computer Networking, A Top-Down Approach, Seventh edition, Pearson,2017 .
2	Behrouz A Forouzan, Data and Communications and Networking, Fifth Edition, McGraw Hill, Indian Edition
Reference Books	
1	Larry L Peterson and Bruce S Davie, Computer Networks, fifth edition, ELSEVIER.
2	Andrew S Tanenbaum, Computer Networks, fifth edition, Pearson

Course Outcomes: At the end of the course, the student will be able to:

CO	Course Outcomes	RBT Level	RBT Level Indicator
CO1	Define and understand the fundamentals of network communication and associated, hardware and software components	U	L1/L2
CO2	Identify various layers of networking and its protocols	Ap	L3
CO3	Analyze the concepts of communication networks	An	L4
CO4	Evaluate different networking concepts, its challenges.	E	L5
CO5	Design a network based on the requirements using modern tools	C	L6

Mapping of Course Outcomes to Program Outcomes:

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2	3													
CO3		3												
CO4			3	2										
CO5					5									

Weblinks and Video Lectures (e-Resources)

1	http://faculty.salina.k-state.edu/tim/ossq/
2	https://www.cs.uic.edu/~jbell/CourseNotes/OperatingSystems/index.html
3	http://www.naturigraw.com/schaum-s-outline-ofoperating-systems.pdf
4	https://www.coursera.org/lecture/technical-support-fundamentals/moduleintroduction-l3n9l

CIE- Continuous Internal Evaluation (50 Marks)

Bloom's Category	Theory			
	Continuous Assessment Tests (IAT)		Continuous Comprehensive Assessment (CCA)	
	IAT-1	IAT-2	CCA-1	CCA-2
	50 Marks	50 Marks	50 Marks	50 Marks
Remember	10	10		
Understand	15	10		
Apply	15	20		
Analyse	10	10		
Evaluate			25	25
Create			25	25

CIE Course Assessment Plan

CO's	Marks Distribution						Total Marks	Weightage
	Test-1			Test-2				
	Module-1	Module-2	Module 2 to 2.5	Module-2.5 to 3	Module-4	Module-5		
CO1	10	10	5	5	10	5	45	45%
CO2	5	5	5	5	5	10	35	35%
CO3		5	5		5	5	20	20%
CO4								
CO5						--	--	
Total	15	20	15	10	20	20		

SEE- Semester End Examination (50 Marks)

Bloom's Category	SEE Marks (90% Theory+10% Practical Questions)
Remember	10% + 2%
Understand	20% + 2%
Apply	40% + 4%
Analyse	20% + 2%
Evaluate	
Create	

SEE Course Plan

CO's	Marks Distribution						Total Marks	Weightage
	Module-1	Module-2	Module 2 to 2.5	Module-2.5 to 3	Module-4	Module-5		
CO1	10	10	5	5	10	10	50	50%
CO2	5	5	5	5	5	5	30	30%
CO3	--	--	5	5	10	--	20	20%
CO4	--	--	--	--	--	--	--	--
CO5	--	--	--	--	--	--	--	--
Total	15	15	15	15	25	15	100	100%